An Unified Machine Learning Pipeline for Early Prediction of Heart Disease, Diabetes Mellitus, and Alzheimer's Disease

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***Abstract*—Due to the increase in the global burden of chronic non-communicable diseases like cardiovascular disease, diabetes mellitus, and Alzheimer's disease, there is a need to create intelligent and scalable clinical decision support systems that can effectively perform early risk stratification. This paper proposes a machine learning-based multidisease prediction model that focuses on three common diseases based on large amounts of structured clinical data. In predicting heart disease, six classifiers including Logistic Regression, K-Nearest Neighbor, Decision Tree, Random Forest, Extra Trees, and XGBoost were used on the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2020 dataset of 319,795 patients where XGBoost was the most accurate with an accuracy of 89.3%, and Logistic Regression had the best recall with an AUC-ROC of 0.84.For the prediction of diabetes, five different classifiers have been evaluated using the dataset of 253,680 records of CDC BRFSS 2015 Diabetes Health Indicators, in which XGBoost classifier produced the highest accuracy of 76.6%, and Random Forest showed better recall of 80.6% along with an F1-score of 73.5%. In the case of predicting Alzheimer's disease, six different classifiers have been trained using 74,284 clinical records, and it is observed that Random Forest has provided 82.47% accuracy along with an AUC-ROC value of 0.8984. Preprocessing pipelines consisting of class balancing through SMOTE technique, StandardScaler normalization, and stratified splitting of the dataset have been used in all modules in order to produce generalized results.**

***Index Terms*—Heart disease prediction, diabetes classification, Alzheimer's disease detection, machine learning, XGBoost, Random Forest, SMOTE, clinical decision support, CDC BRFSS**

I. Introduction

One person passes away every eleven seconds due to diabetes and/or its complications in any part of the world. Another one dies every thirty-three seconds because of cardiovascular diseases. One new case of dementia is reported every three seconds throughout the world – most of which is due to Alzheimer's disease. They are not just numbers that can be found in any annual report on healthcare, but the reality that is taking place quietly in clinics, homes, hospitals in every country, in every social class, and among all ages. According to the WHO, approximately 74% of deaths per year in the world are due to NCDs (non-communicable diseases) which include cardiovascular diseases, diabetes mellitus, and Alzheimer's disease.

It is impossible to disassociate the human toll of this situation from an institutional failure to make timely and intelligent diagnoses. Take the case of a 54-year-old man whose relatively minor symptoms of chest pain are overlooked in an underfunded clinic until he experiences a massive cardiac incident three weeks down the line. Or take the story of a middle-aged woman whose persistent thirst and blurring of her vision are put down to stress until, years later, she is diagnosed with uncontrolled type 2 diabetes after her organs have already incurred some damage. Or the story of a granddad who has forgotten where he left his glasses but is not assessed until his Alzheimer's disease is far too progressed to be treated.

This problem is not only clinical in nature; it is computational and informational. A large amount of structured data is collected each day through regular health questionnaires, electronic medical records, and surveillance efforts at a population level. However, this data is mostly left underutilized and never really used to gain valuable diagnostic insights for individual patients. It is here that the value of machine learning (ML) becomes clear. In recent years, ML-based classifiers have demonstrated strong predictive capability across several medical domains, including cardiovascular risk stratification [1], [2], diabetes detection [3], and cognitive decline screening [4].

However, a consistent weakness in the current body of research is the isolationism inherent in each solution, where each model is trained separately, tested on one specific disease, and rarely built to operate in an integrated and extendable framework [5]. Furthermore, accuracy alone is insufficient to drive adoption of a solution by a doctor. It is impossible, and even unethical, to expect doctors to take predictions at face value from models without an explanation. In the critical world of healthcare, interpretability is a necessity, not an option.

Driven by the convergence of the above requirements of early disease detection, multiple disease detection, interpretability, and scalability of deployment, the current paper puts forward a structured machine learning framework that

can be used to predict the existence of any one of the three highly prevalent chronic diseases: heart disease, diabetes mellitus, and Alzheimer's disease. Each disease component is separately trained using disease-specific data pre-processing, class balance using SMOTE, selective feature extraction, and multiple metric evaluation.

The principal contributions of this work are as follows:

- A novel multidisease prediction framework is introduced that tackles heart diseases, diabetes mellitus, and Alzheimer's disease using disease-specific but architecturally unified machine learning models applied to large publicly available clinical datasets.
- A systematic benchmarking experiment is performed for all three diseases where multiple classifiers such as Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, and Multi-Layer Perceptron are compared using the same evaluation metrics in order to choose the most efficient predictor for each disease.
- SMOTE oversampling is used only on training splits to overcome class distribution imbalance without causing data leakage and thus enhancing the ability of the model to predict minority classes which represent positive disease cases.
- Optimization of the threshold using the validation set according to the F1-Score is used to optimize the precision recall trade-off, making it possible to adjust the model to increase the sensitivity in scenarios where the diagnosis is more important than false positives.

II. Related Works

Yaganteeswarudu [6] has developed a multi-disease prediction system that employs machine learning, TensorFlow, and a Flask API to predict various diseases like diabetes, diabetic retinopathy, heart disease, and breast cancer in one platform. This model utilizes relevant clinical parameters and stores its training models using Python Pickle.

Acharya [7] did a comparative study on machine learning algorithms for heart disease prediction using UCI Heart Disease dataset. It compared the performance of KNN, SVM, Naïve Bayes, AdaBoost, Random Forest, and Artificial Neural Network based on various evaluation parameters such as accuracy, precision, recall, and F1-score. Feature selection, grid search, and probability calibration were some of the techniques used to fine-tune the models. It was found that Artificial Neural Network and SVM had the highest performance on this dataset.

Islam et al. [8] conducted an exhaustive review on the use of machine learning algorithms in predicting chronic diseases including cardiovascular disease, diabetes, cancer, liver diseases, and neurological diseases. The analysis involved various machine learning and deep learning algorithms such as SVM, Random Forest, Decision Tree, and Naïve Bayes. Feature selection techniques were also explored including LASSO, mRMR, and RELIEF. This work shows the possible use of ML algorithms in improving disease prediction in clinical settings.

Mujumdar and Vaidehi [9] have introduced a machine learning-based diabetes prediction model that integrates conventional clinical factors (such as glucose, BMI, age, and insulin), as well as additional factors related to diabetes. The aim of the research was to develop a better classification algorithm using a better dataset and pipeline-based prediction model. The results revealed improved diabetes classification performance in comparison with other existing algorithms.

Shukla et al. [10] provided a survey of approaches to the detection of Alzheimer's disease using automatic pipelines and machine learning algorithms. In the course of the survey, approaches related to biomarkers, multimodal data fusion, and image registration were considered. It was revealed that ML-based automatic pipelines can reach up to 95% classification accuracy in both single- and binary-class classification tasks, whereas multiclass classification and distinguishing Alzheimer's disease from MCI and its variants is more problematic. The necessity of the application of multimodal approaches in such problems was also emphasized.

Bardia and Sophiya [11] analyzed the predictive capacity of various machine learning models in terms of diabetes diagnosis using the RTML and PIMA Indian datasets. Mutual information was used for the feature selection, and SMOTE for solving the problem of class imbalance. Several models were evaluated including Random Forest, Logistic Regression, KNN, Neural Networks, AdaBoost, and XGBoost. The best results were reached by XGBoost model: it provided 83.22% accuracy, 0.86 AUC, and 85% cross-validation score.

Ganguly et al. [12] performed a comparative evaluation of six machine learning approaches for cervical cancer risk classification using an 838-instance database with 36 features. The results showed that the Support Vector Machine (SVM) produced the best result with an accuracy of 99.60%, showcasing the efficacy of machine learning algorithms for accurate cervical cancer risk assessment and early detection.

Aluru et al. [13] developed a machine learning algorithm for detecting Parkinson's disease in its early stages by using features of human voice as biomarkers. The authors used such classifiers as Naïve Bayes, SVM, KNN, and Decision Tree, as well as deep learning algorithms like CNN, to detect individuals with high risks of having Parkinson's disease.

Chavda and Sampath [14] developed a framework based on Convolutional Neural Networks (CNN) for detecting Alzheimer's Dementia with the help of MRI clinical data. The proposed framework was able to get 98.64% accuracy during the training stage and 96% accuracy during the testing stage and proved better than other models like VGG16, VGG19, and InceptionResNetV2. This study proves the usefulness of CNN in detecting the presence of Alzheimer's Disease from MRI data.

Kodati Bhanusri et al. [15] have suggested a machine learning approach for heart stroke prediction with the help of models like XGBoost, Decision Tree, KNN, SVC, and Random Forest. These machine learning models were validated by the accuracy, precision, recall, and F1-score metrics. Further, SHAP-based XAI is used for the detection of the most

significant medical features in predictions.

It can be seen that all the reviewed research papers show the efficiency of machine learning and deep learning approaches in predicting chronic diseases like diabetes, heart diseases, Alzheimer's disease, Parkinson's disease, and heart stroke. Approaches like SVM, random forest, XGBoost, CNN, ensemble learning, SMOTE, and Explainable AI have been quite successful in this field.

III. Methodology

A. System Architecture

The architecture of the proposed system is designed in an end-to-end machine learning pipeline that consists of three modules: one per each disease under consideration, such as heart disease, diabetes mellitus, and Alzheimer's disease, where each of these modules follows the same approach in preprocessing, evaluation and explanation. When the user accesses the system, he chooses the corresponding module for the disease he is interested in, whereafter the structured clinical data in a tabular format goes through the preprocessing step, where missing data is filled, duplicate entries are dropped, categorical variables are encoded, and numerical features are normalized. Then the preprocessed features go to the corresponding classification module for the disease, where various machine learning models for classification are built and evaluated.

B. Dataset Description

The heart disease dataset was taken from the BRFSS survey held in 2020 and available on Kaggle, and comprises 319,795 observations and 18 features, which are related to demographical characteristics (age, sex, and race), lifestyle aspects (such as smoking, drinking, sports activities and hours of sleep), clinical variables (BMI, physical and mental health days), and information about comorbidities such as stroke, asthma, kidney disease, skin cancer, and diabetes.The binary dependent variable HeartDisease indicates whether the patient has heart disease or not.

The diabetes data set is collected from the Diabetes Health Indicators 2015 CDC BRFSS data set that is also available on Kaggle and contains 253,680 survey responses in 21 variables, including high blood pressure, high cholesterol, BMI, smoking, stroke, physical activity, general health, and demographics like gender, age, education, and income. The binary target variable called 'Diabetes binary' separates diabetics from non-diabetics.

The Alzheimer's disease data set is extracted from Kaggle and contains 74,284 observations and 25 variables, which include demographics like age, gender, and education; lifestyle variables, like physical activity, smoking, and alcohol; clinical parameters like BMI, blood pressure, and cholesterol; neurology variables, like cognitive scores and depression; and socioeconomic variables, like employment, income, and stress.

C. Data Preprocessing

There are no missing values for both the Heart Disease and Diabetes datasets as proven by the null-value analysis conducted. Missing numerical attributes in the Alzheimer's dataset are handled using median imputation while missing categorical attributes using mode imputation to ensure consistent distribution of features. In the case of the Diabetes dataset, duplicates are identified and removed using the drop duplicates function in the pandas package.

Label encoding is used to encode binary categorical variables such as HeartDisease, Smoking, Stroke, Sex and other comorbidity flags in the Heart Disease dataset as Yes/No $\rightarrow$ 1/0 and Male/Female $\rightarrow$ 1/0. One-Hot Encoding is used to transform multi-class variables such as AgeCategory, Race, GenHealth and Diabetic in the Heart Disease Dataset using the ColumnTransformer class in scikit-learn. The attribute Diabetic is encoded such that "No, borderline diabetes" maps to 0 and "Yes (during pregnancy)" maps to 1.

All feature matrices are standardized using the StandardScaler in scikit-learn resulting in features that have zero mean and unit variance. All the three datasets are partitioned into train, validation, and test sets using stratified sampling.In the case of Heart Disease and Diabetes datasets, the Synthetic Minority Over-sampling Technique (SMOTE) was only applied on the training dataset to counteract class imbalance without causing any data leakage to the validation and testing sets.

D. Feature Engineering

Feature engineering was mainly focused on the Alzheimer's disease dataset. The CognitiveTestScore variable was converted into four risk categories, which include Very High Risk, High Risk, Moderate Risk, and Low Risk categories using the pd.cut() method with the ranges of [0, 40, 60, 75, 100] in order to help the models to learn non-linear threshold-based relationship between cognitive ability and disease progression. A second-degree polynomial feature Cognitive Score Squared was created using square of the raw cognitive test score for learning non-linear relationship between cognitive abilities and risk of Alzheimer's. Additionally, interaction features involving age with other variables were generated to learn the effects of the aging process in combination with other factors.

For the Heart Disease and Diabetes datasets, feature selection was performed based on exploratory data analysis, correlation heatmap, and model-based feature importance assessment. All features in their original form were used during the training phase to let the ensemble models select important features internally.

E. Model Development

For predicting heart disease, six classifiers were trained with the preprocessed data, and those classifiers are Logistic Regression, K-Nearest Neighbor (KNN), Decision Tree, Random Forest, Extra Trees, and XGBoost. Training of models were done in the same way with respect to preprocessing of data in order to make a fair and proper comparison between all models by tuning thresholds on validation set.

For predicting diabetes, five classifiers were considered, and those are Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost, all of which are trained by following the same procedure for removing duplicates, feature scaling, SMOTE, and stratified splitting.

For predicting Alzheimer's disease, six classifiers were used, and those classifiers are Logistic Regression, Random Forest, Gradient Boosting, XGBoost, LightGBM, and Multi Layer Perceptron (MLP) neural network classifier. Also, Voting Classifier with soft voting mechanism was formed from selected base estimators. To prevent overfitting in a high dimensional feature space, Stratified K-Fold cross-validation with five folds was performed using the cross val score method of scikit-learn.

F. Evaluation Metrics

The evaluation metrics adopted in the framework were standardized across all three disease models. Accuracy provides information about the overall predictive performance of the model; however, it should be considered alongside more sensitive metrics due to class imbalance in the datasets. Precision measures the proportion of positive predictions that are actually positive, while recall measures the proportion of positive samples correctly identified by the model. Recall is particularly important in disease screening, as missed diagnoses may have serious clinical implications. The F1-Score provides a balanced measure of precision and recall, making it suitable for imbalanced datasets. AUC-ROC evaluates the model's ability to distinguish between classes across different classification thresholds. Confusion matrices, plotted using Seaborn heatmaps, were used for all selected models to provide class-wise classification results. For the Alzheimer's dataset, Matthews Correlation Coefficient (MCC) and Cohen's Kappa were also calculated.

IV. Results and Discussion

A. Heart Disease Results

The module of heart disease prediction was tested using six classifiers such as Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Extra Trees, and XGBoost on CDC BRFSS Heart Disease dataset which includes 319,795 patients with demographic, lifestyle, and medical characteristics. One-Hot Encoding and binary mapping were used to preprocess categorical data; SMOTE technique was applied to address class imbalance issue; the StandardScaler technique was used to scale the features. The metrics such as Accuracy, Precision, Recall, F1-Score, and AUC-ROC were applied for evaluating each classifier.

Table I summarizes the comparative performance of all classifiers.

The results do not indicate the superiority of one classifier over others on all metrics. XGBoost had the highest accuracy (89.3%) and precision (35.8%), which indicates that this algorithm performs well on classification tasks. Extra Trees performed best on F1-Score (37.4%), which demonstrates good performance with respect to both precision and recall. Logistic Regression, although having worse accuracy, performed best with respect to recall (78.4%) and AUC-ROC (0.84). This result shows that in healthcare applications it is crucial to consider multiple metrics.

TABLE I
Performance Metrics for Heart Disease Prediction Models

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	74.6%	22.2%	78.4%	34.6%	0.84
K-Nearest Neighbors (KNN)	84.7%	26.5%	44.1%	33.1%	0.76
Decision Tree	82.9%	23.9%	45.9%	31.4%	0.80
Random Forest	86.2%	30.6%	47.9%	37.3%	0.82
Extra Trees Classifier	81.3%	26.2%	65.1%	37.4%	0.83
XGBoost	89.3%	35.8%	31.1%	33.3%	0.82

Considering the highest values of recall and AUC-ROC, Logistic Regression was chosen as a primary classifier for further consideration. In screening context, the most important metric is the recall value, since it represents the ability of the algorithm to detect positive instances. High recall leads to low numbers of false negatives and therefore lowers the possibility of missed diagnostics and early intervention. Fig. 1 illustrates the ROC curve comparison between all classifiers.

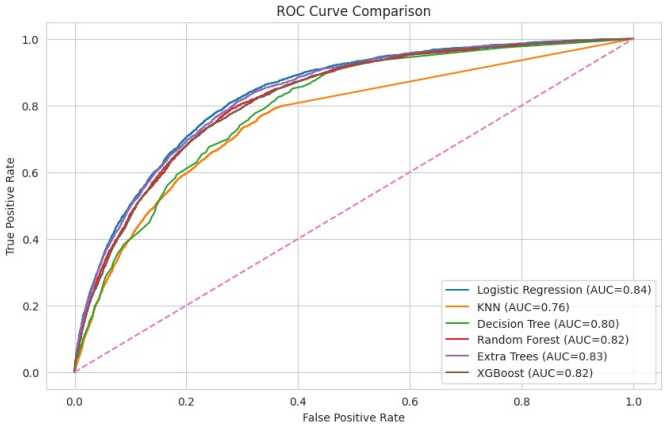


Fig. 1. ROC Curve Comparison of Machine Learning Models for Heart Disease Prediction

The model with the highest AUC-ROC was Logistic Regression, showing it to have the highest discriminating power. Ensemble learning methods, including Random Forest, Extra Trees, and XGBoost also showed good results, although all were above the line threshold and therefore showed their superior ability to perform better than purely random classification. The confusion matrix for the selected Logistic Regression model is displayed in Fig. 2.

From the confusion matrix, it is clear that Logistic Regression managed to identify a majority of the heart diseases in true positives. Even though this comes at the expense of a higher false positive rate compared to some ensemble classifiers, it is still a trade-off worth making as the cost of missing the risky patients is much more dangerous than conducting further checks in healthy patients. Overall, the heart disease module proved itself capable of cardiovascular risk prediction in a clinically meaningful manner, with Logistic Regression being the best model for this task.

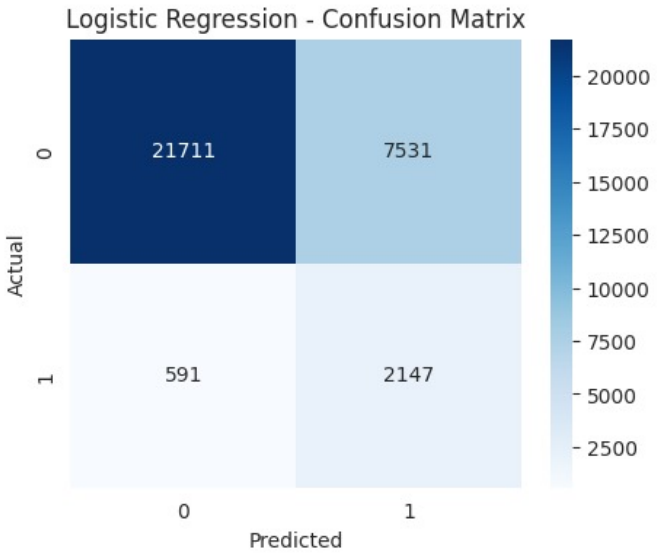


Fig. 2. Confusion Matrix of the Logistic Regression Model for Heart Disease Prediction

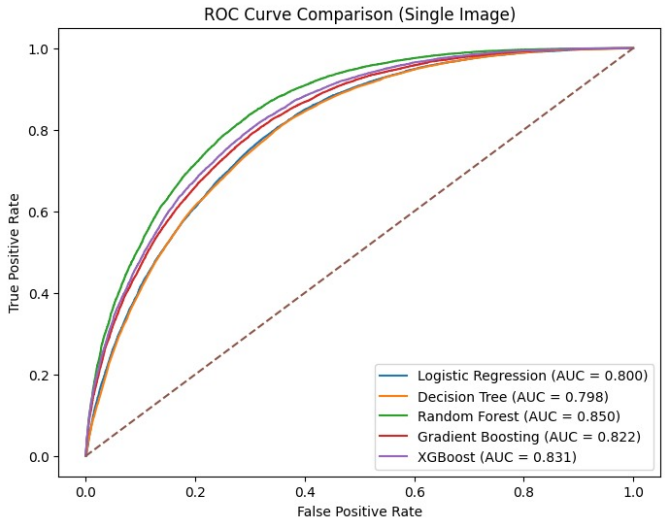


Fig. 3. ROC Curve Comparison of Machine Learning Models for Diabetes Prediction

B. Diabetes Results

The diabetes prediction module was tested on the CDC BRFSS 2015 Diabetes dataset which includes 253,680 instances containing demographic, lifestyle, and clinical features. Preprocessing steps included the elimination of duplicate instances, correction of class imbalance through SMOTE technique, exploration of the dataset, correlation analysis, and normalization using StandardScaler. Five machine learning models were employed for evaluation including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost.

Table II summarizes the comparative performance of all evaluated classifiers.

TABLE II
PERFORMANCE METRICS FOR DIABETES PREDICTION MODELS

Machine Learning Model	Accuracy (%)	Precision	Recall	F1-Score
Logistic Regression	71.9%	63.7%	74.0%	68.5%
Decision Tree	71.2%	61.6%	79.5%	69.4%
Random Forest	76.0%	67.5%	80.6%	73.5%
Gradient Boosting	74.0%	69.1%	66.8%	67.9%
XGBoost	76.6%	69.6%	67.9%	68.8%

The results show that ensemble approaches outperformed single classifiers on the majority of the metrics. XGBoost has the highest accuracy of 76.6%, while Random Forest shows the highest recall (80.6%) and F1-Score (73.5%), indicating its ability to detect diabetic patients without compromising the balance between precision and recall. Logistic Regression showed stable output values, while Decision Tree has relatively high recall. Generally speaking, ensemble classifiers were better in capturing interactions in the data.

For detailed analysis of the results, we chose the Random Forest classifier due to its high recall and the best F1-Score. In the case of healthcare screening tasks, recall is the most important metric since it reflects the ability of the classifier to recognize true positives. Thus, the higher recall leads to the reduction of false negative results and minimizes the risk of misdiagnosing patients. The high F1-Score indicates that the increase of the sensitivity did not lead to the significant loss of precision. Fig. 3 presents the ROC curve comparison across all evaluated classifiers.

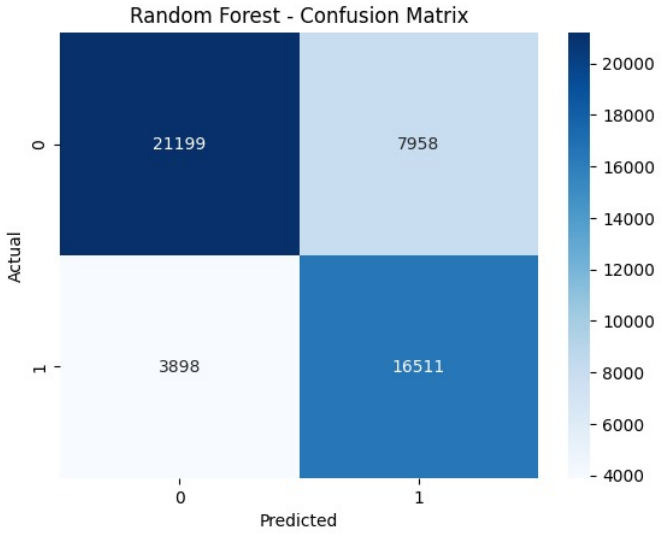


Fig. 4. Confusion Matrix of the Random Forest Model for Diabetes Prediction

All classifiers performed above the diagonal line showing good classification results beyond the random guessing line. Models based on ensembling techniques showed high discrimination power irrespective of the classification threshold chosen thus making them more suitable for separating patients with diabetes from non-diabetic subjects. The confusion matrix for the chosen Random Forest algorithm is provided in Fig. 4.

According to the confusion matrix, the Random Forest algorithm successfully classified a considerable number of people who have diabetes while providing classification results of similar quality for both classes. The small number of false negatives indicates the high sensitivity of the model which is an important property when diagnosing diseases in healthcare because delays in treatment can cause many undesirable consequences in future.

C. Alzheimer's Disease Results

A whole array of machine learning algorithms , such as Logistic Regression, Multi Layer Perceptron Neural Network, Random Forest, XGBoost and LightGBM, along with Gradient Boosting were applied as means of evaluation of Alzheimer's disease predictor model. Having done that, the dataset was found to consist of 74,283 samples and 69 features altogether, not to mention other stuff that I did not really count. Then, we divided the dataset into three parts, namely Training set, Validation set and Testing set, comprising 51,998, 18,941 and 18,942 instances respectively.

Table III, in general terms, provides overall performance of the model via Test Accuracy, AUC-ROC, F1-Score and Recall, as can be seen. Eventually, the best performance among all of those algorithms was provided by Random Forest classifier rather than the other ones. The figures of that algorithm's performance include test accuracy 82.47%, AUC-ROC 0.8984, F1-Score 0.7955, and Recall 0.8200. This particular figure of AUC-ROC is rather noticeable and proves the good separating power of the classifier, or in other words – discrimination of the classes.

TABLE III
PERFORMANCE COMPARISON OF ALZHEIMER'S DISEASE PREDICTION MODELS

Model	Accuracy	AUC-ROC	F1-Score	Recall
Logistic Regression	72.26%	0.8041	0.7003	0.7795
MLP Neural Network	72.82%	0.8060	0.6656	0.6509
Random Forest	82.47%	0.8984	0.7955	0.8200
XGBoost	76.95%	0.8473	0.7219	0.7198
LightGBM	75.21%	0.8414	0.7282	0.7986
Gradient Boosting	75.06%	0.8287	0.7022	0.7075

However, both Logistic Regression and MLP neural network models performed relatively poorly with their accuracy being approximately 72.26% and 72.82% respectively. Although they have been providing rather good classification results, yet they could not provide predictions in a really confident manner as compared to the tree based learners which were created using ensembles. It seemed that even XGBoost and LightGBM models were also somewhat moderate with their accuracy range of 75% - 77%.

The really high efficiency of the Random Forest model seems to be associated with its capability to identify nonlinear relationships among various attributes such as demographic, clinical, cognitive, and lifestyle. In other words, it is able to recognize non-linear correlations between features despite some interlacement and mixing of their signals.

Fig. 5 shows an example of the confusion matrix of the Random Forest classifier. Generally, the classifier predicted that 9,165 people were free of Alzheimer's disease. At the same time, it classified 6,457 people as patients with the disease. However, 1,903 people were misclassified into the group of not having the disease. At the same time, 1,417 individuals were incorrectly classified as having Alzheimer's disease. In most cases, as the false negatives are fewer, the algorithm is very good at detecting those people who have the disease.

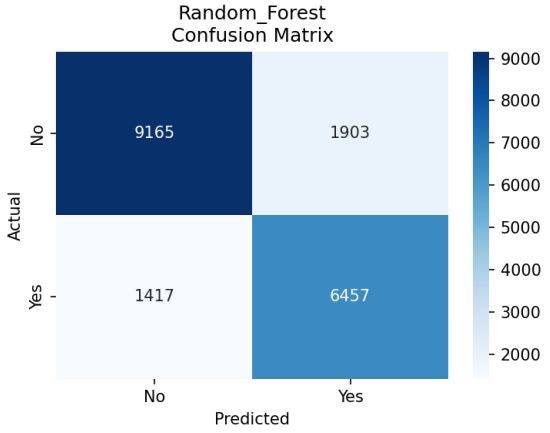


Fig. 5. Confusion Matrix of the Random Forest Model for Alzheimer's Disease Prediction

The ROC plot for the Random Forest Classifier is shown in Fig. 6. Generally speaking, the ROC plot is located rather far from the dashed diagonal line, which makes the classifier seem to be highly discriminative at different threshold levels. Moreover, the AUC equals 0.8984, which appears to be a rather good indicator of high class separability.

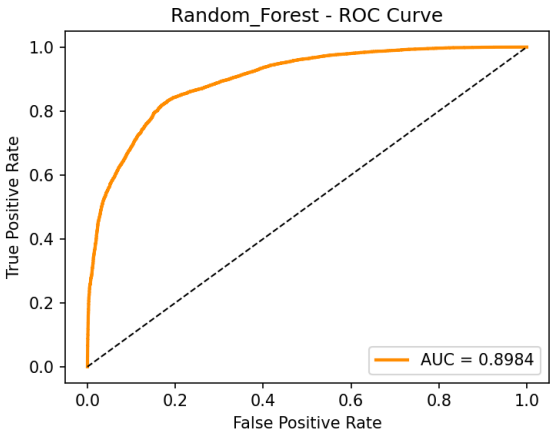


Fig. 6. ROC Curve of the Random Forest Model

According to the classification report, it can be observed that the Random Forest Model performed rather similarly for both types of data, although not exactly the same, but

nearly equal results. Thus, for non-Alzheimer's cases, the precision equals 0.87, while the recall equals 0.83, while for Alzheimer's patients, the precision and recall equal 0.77 and 0.82, respectively. It means that the F1 score of 0.83 can be considered as reliable due to equilibrium of the two classes.

V. Conclusion and Future Work

In this paper, MediSign, a comprehensive machine learning framework for the early prediction of three high-prevalence chronic diseases (heart disease, diabetes mellitus, and Alzheimer's disease) was described. Disease modules were individually developed through their own respective data processing pipelines, class balancing using SMOTE and extensive classification using multiple metrics and several classifiers. The experimental analysis showed a highly accurate prediction for each disease module: Logistic Regression achieved the highest recall (78.4%) and AUC-ROC (0.84) for heart disease; Random Forest demonstrated the highest recall (80.6%) and F1-Score (73.5%) for the prediction of diabetes mellitus; finally, Random Forest once again proved to be the best performing algorithm for detecting Alzheimer's disease with an accuracy of 82.47% and AUC-ROC of 0.8984. All these results indicate the applicability of ensemble-based classification methods in conjunction with proper preprocessing and evaluation techniques on structurally different disease areas.

Although the above-mentioned results are highly promising, there are some limitations to the current framework. First of all, all used datasets are based on surveys and are of cross-sectional type. Also, none of the disease modules currently interact with each other while being classified separately.Moreover, the relatively low precision obtained from the heart disease module clearly illustrates the need for better learning techniques for the minority classes apart from SMOTE.

The future directions of this research include multiple aspects that will improve these limitations. The integration of SHAP and LIME explainability layers will be included to provide clear explanation for clinicians to understand the reasons behind each prediction. Future work will include longitudinal patient data as well as imaging and electronic health records to enhance the predictive context. Additionally, development of a risk score dashboard that incorporates all three disease modules and their predictions in one place for a clinical decision support will be done.

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